

Students can correctly identify and name points, lines, segments, rays, and planes.

Do Now

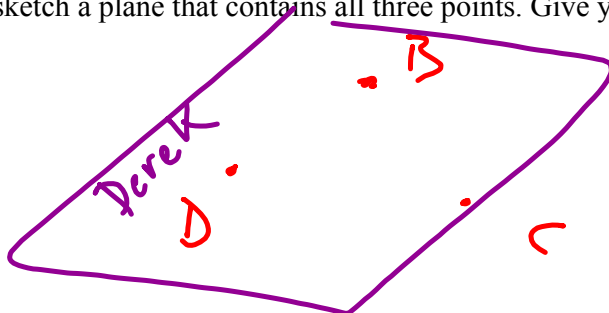
- 1) Place your name tent on your desk.
- 2) a. What does it mean for two points to be collinear?

"On same line"

- b. What does it mean for two points to be coplanar?

"on same plane"

- c. Sketch three points that are not collinear. Give your points names. Then sketch a plane that contains all three points. Give your plane a name.



- 3) a. Name three points that are collinear.

R, Q, S

- b. Name a fourth point that is not collinear points in part a.

T or W

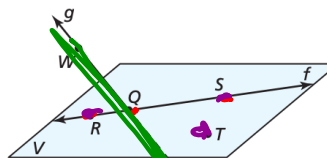
- c. Name a point that is coplanar with points R, Q, and T.

S

- d. Name a point that is not coplanar with points R, T, and S.

W

- e. Draw and label a new point, point J, that is collinear with points W and Q.



Naming Conventions <i>lower case!</i>		
Line	A straight path that extends forever in both directions Note: $\overleftrightarrow{AB}$ same $\overleftrightarrow{BA}$	line ℓ
Line Segment	A straight path connecting 2 endpoints. (Does not go forever) Note: $\overline{AB}$ same $\overline{BA}$	
Ray	A straight path extending in one direction forever from endpoint. Note: $\overrightarrow{AB} \neq \overrightarrow{BA}$	
Opposite Rays	2 rays that share an endpoint and extend in opposite directions. $\overrightarrow{CB}$ is opp to $\overrightarrow{CA}$	

With rays, the first letter is *always* the endpoint. The ray starts there and goes in the direction of the second letter.
(if arrow is $\longrightarrow$)
 $\overrightarrow{AB}$
Start

Ex 1: Name each of the following in as many different ways as possible:

a.

line m , $\overleftrightarrow{GP}$, $\overleftrightarrow{PG}$

b.

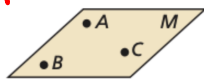
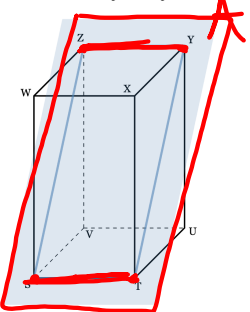
$\overline{VU}$ $\overline{UV}$

c.

$\overline{ZR}$ $\overline{RZ}$
 ~~$\overline{NR}$~~
Also correct but less common:
 $\overline{ZNR}$, $\overline{RNZ}$

d.

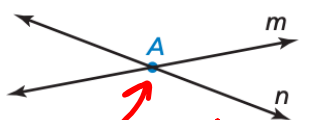
$\overrightarrow{GJ}$
 $\overrightarrow{JG}$
 $\overrightarrow{GFJ}$ ✓
 $\overrightarrow{JFG}$
 $\overrightarrow{GF}$

Naming Planes <i>capital letter</i>	
Plane	A flat surface extending forever in all directions. Plane M  Plane ABC Plane BCA Plane CAB
Ex 2: Name the plane in the highlighted diagram in as many ways as possible. Plane STZY Plane A Plane SYZ Plane YZT  (any 3 of the 4 points on plane)	

Intersections

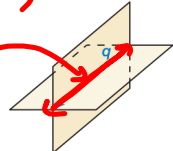
An **intersection** of two or more geometric figures is the set of points that the figures have in common.

Ex 3a.: The intersection of line m and line n shown below is Point A.

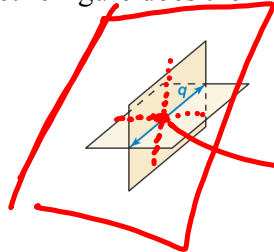


point A (intersection)

b.: The intersection of two planes is a line



c.: What geometric figure does the intersection of three planes make?

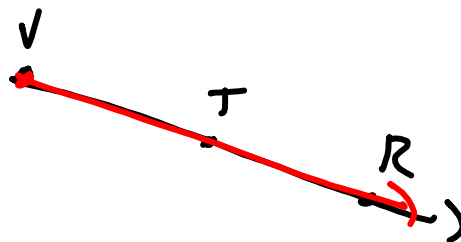


A point

Practice:

Complete

- DeltaMath 1.2 CheckIn
- Assignment 1.2 on paper



$\overrightarrow{VR}$

$\overrightarrow{VT}$

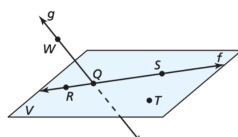


Geometry
Fletcher

Name _____
Date _____

Assignment 1.2: Naming Conventions

- Use the diagram below to answer the following questions:



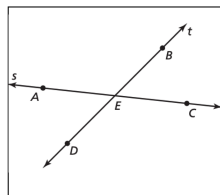
Give two other names for $\overleftrightarrow{WQ}$.

Give another name for plane V.

Name three points that are collinear. Then name a fourth point that is not collinear with these three points.

Name a point that is not coplanar with R, S, and T.

- Use the diagram to answer the following questions.



What is another name for $\overleftrightarrow{BD}$?

What is another name for $\overleftrightarrow{AC}$?

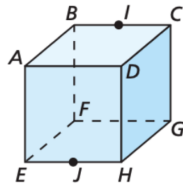
What is another name for $\overleftrightarrow{AE}$?

Name all rays with endpoint E.

Name two pairs of opposite rays.

Name one pair of rays that are not opposite rays.

3. Use the diagram to answer the following questions.



- Name a point that is collinear with points E and H .
- Name a point that is collinear with points B and I .
- Name a point that is not collinear with points E and H .
- Name a point that is not collinear with points B and I .
- Name a point that is coplanar with points D , A and B .
- Name a point that is coplanar with points C , G , and F .
- Name the intersection of plane AEH and plane FBE .
- Name the intersection of plane BGF and plane HDG .